Questions

Q1.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 4 shows two lenses, P and Q, arranged to form a simple telescope.

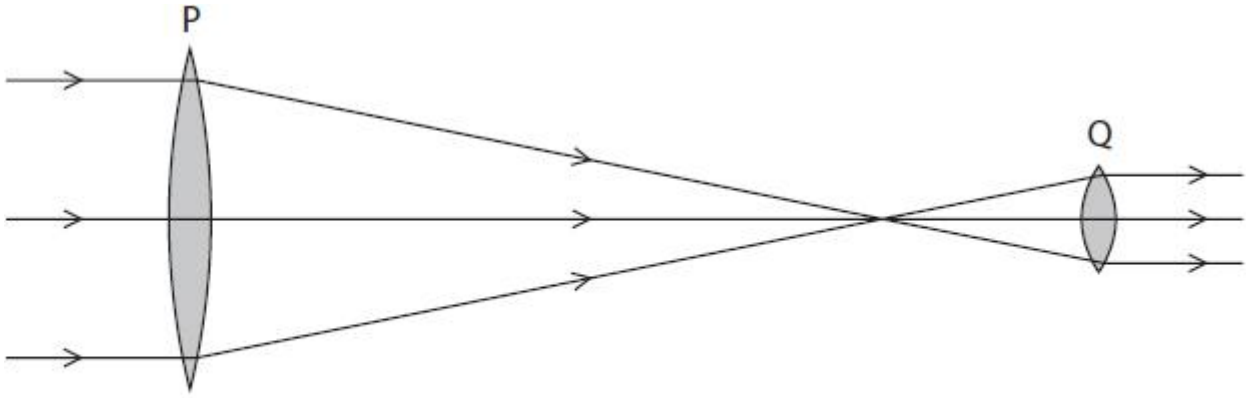

Figure 4

(i) State **one** use for a telescope.

(1)

.....

.....

(ii) Which of these describes the lenses?

(1)

☐ **A** P is converging, Q is diverging
☐ **B** P is diverging, Q is converging
☐ **C** P and Q are both converging
☐ **D** P and Q are both diverging

(iii) The focal length of lens Q is 0.14 m.

Calculate the power of the lens.

(2)

Use the equation

$$\text{power} = \frac{1}{\text{focal length in m}}$$

power of lens = ... dioptre

(Total for question = 5 marks)

Q2.

Figure 8 shows two types of force that act in the Sun.

One type of force acts inwards, towards the centre of the Sun.

The other type of force acts outwards, away from the centre of the Sun.

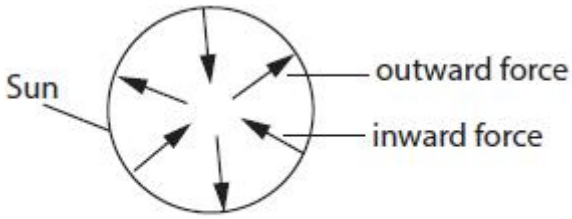

Figure 8

Outward forces are a result of nuclear fusion in the Sun.

State what causes the inward forces.

(1)

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(Total for question = 1 mark)

Q3.

During the evolution of a star, the nebula collapses and becomes a main sequence star.

(i) State what causes the nebula to collapse.

(1)

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(ii) Explain why the nebula stops collapsing as it becomes a main sequence star.

(3)

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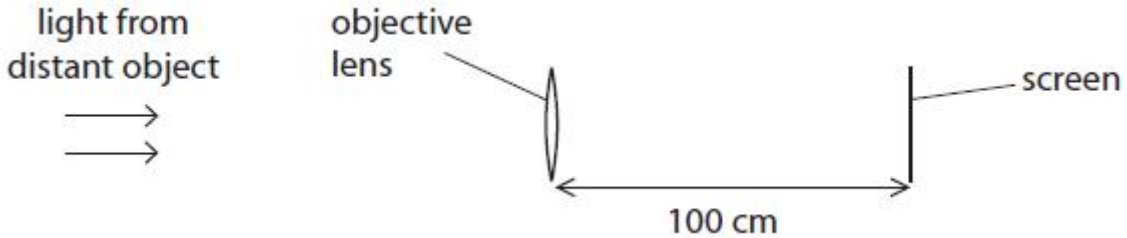
(Total for question = 4 marks)

Q4.

(a) This picture shows Galileo holding a telescope.

A student starts to make a similar telescope.

She places the objective lens near a screen to form a sharp image of a distant object.



(i) Use words from the box to complete the sentence.

(2)

bigger than the object	the right way up	the same size as the object
smaller than the object	upside down	

The sharp image formed is

..... and .....

(ii) State how you can tell that the image formed is real.

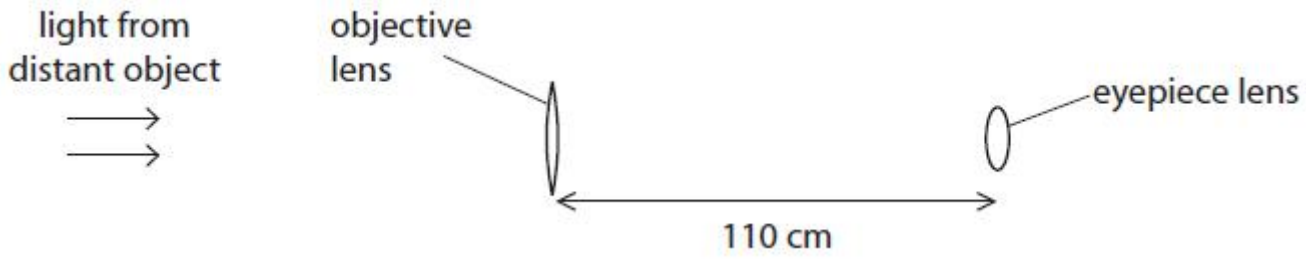
(1)

.....

.....

(iii) She then removes the screen and replaces it with an eyepiece lens.
She moves the eyepiece lens until she can see a sharp, magnified image through it.

The diagram shows the position of the lenses.



The lenses are now separated by 110 cm.
Complete the sentence by putting a cross (☒) in the box next to your answer.
The focal length of the eyepiece is about

(1)

☐ **A** 10 cm
☐ **B** 55 cm
☐ **C** 90 cm
☐ **D** 100 cm

(iv) Complete the sentence by putting a cross (☒) in the box next to your answer.
At the eyepiece of this telescope there is

(1)

☐ **A** refraction by a converging lens
☐ **B** refraction by a converging mirror
☐ **C** reflection by a converging lens
☐ **D** reflection by a converging mirror

(b) State how the invention of the telescope improved the way scientists observed the Universe.

(1)

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.....

(c) The invention of the telescope helped to change ideas about the Solar System.
The heliocentric model replaced the geocentric model.
Describe the difference between the geocentric model and the heliocentric model.

(2)

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(Total for Question = 8 marks)

Q5.

(i) The chart shows the electromagnetic (EM) spectrum.
Some parts of the spectrum have been labelled.

radio	**P**	**Q**	visible light	**R**	**S**	gamma rays

State the name of part **Q**.

(1)

.....

(ii) Three telescopes using different parts of the spectrum have been added to the chart.

radio	**P**	**Q**	visible light	**R**	**S**	gamma rays
Jodrell Bank			Hubble			Compton

Jodrell Bank is located near Manchester, whereas Hubble and Compton are in space.

Explain why some telescopes are located outside the Earth's atmosphere.

(2)

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Q6.

The Sun is a star that is in a stable state called the main sequence.

The Sun, in its main sequence, radiates energy continuously but stays the same size.

(i) State the energy source in the Sun.

(1)

.....

(ii) Name the force that prevents the Sun from expanding.

(1)

.....

(Total for question = 2 marks)

Q7.

During the twentieth century red-shift and CMB radiation were discovered.
They have provided scientists with data to test theories of the origin of the Universe.
Stars have different stages in their evolution.
(i) Which of these gives the next stages in the evolution of the Sun?
Put a cross (☒) in the box next to your answer.

(1)

☐ **A** white dwarf then black hole

☐ **B** neutron star then white dwarf

☐ **C** red giant then supernova

☐ **D** red giant then white dwarf

(ii) Modern telescopes can provide us with more data than the telescopes used 100 years ago. Explain what additional data can be collected and processed using modern telescopes.

(2)

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Q8.

Figure 2 shows part of the circular orbit of a planet around the Sun.

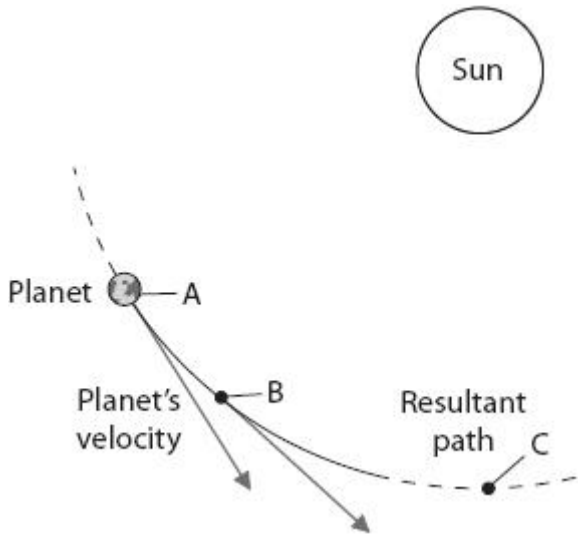


Figure 2

(i) Give the name of the force that keeps the planet moving in a circular orbit.

(1)

.....

(ii) The direction of the velocity of the planet at points A and B is shown by the arrowed lines.

Add an arrowed line to the diagram to show the direction of the velocity at C.

(1)

(Total for question = 2 marks)

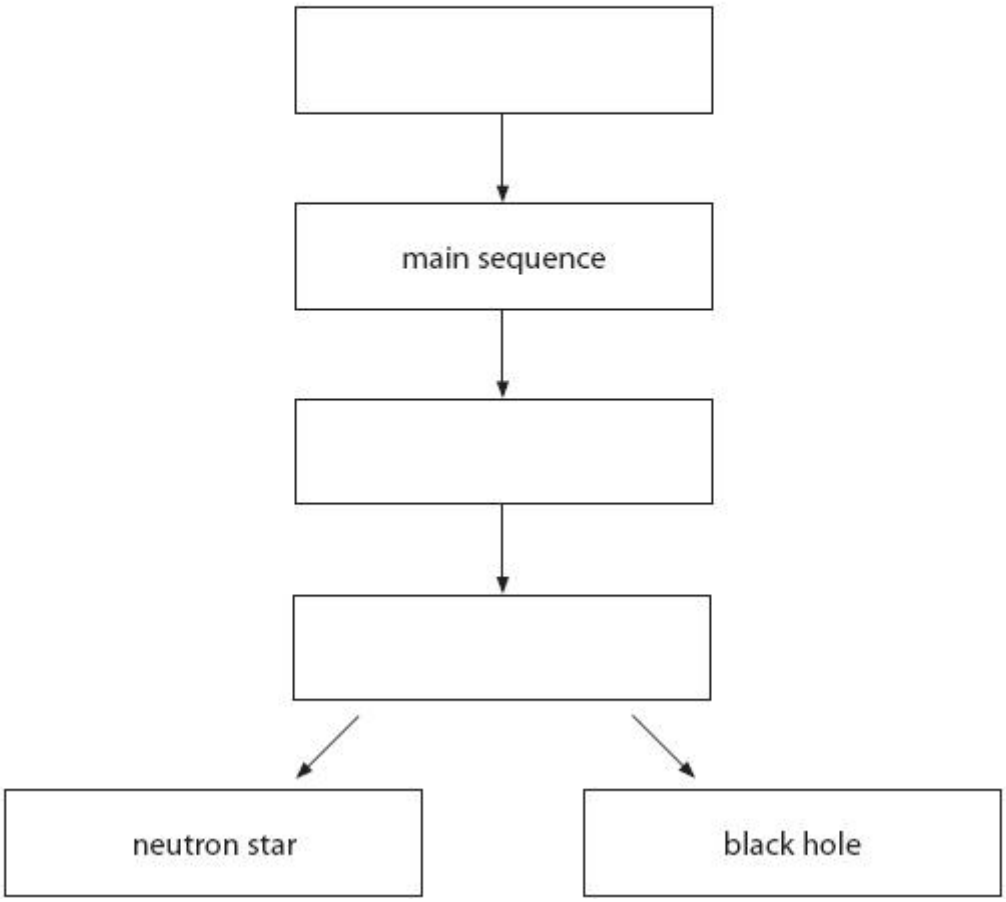
Q9.

Stars that have a mass much larger than that of the Sun have a different life cycle.

Choose words from the box to complete the life cycle of a massive star.

(3)

black dwarf	galaxy	nebula	red dwarf
red super giant	supernova	white dwarf	



(Total for question = 3 marks)

Q10.

The Moon that orbits the Earth is

(1)

- ☐ **A** a comet
- ☐ **B** a dwarf planet
- ☐ **C** a natural satellite
- ☐ **D** an asteroid

(Total for question = 1 mark)

Q11.

(a) Which row of the table shows these objects in the correct order of size?
Put a cross (☒) in the box next to your answer.

(1)

%%%	**smallest** ⟶ **biggest**		
☐ **A**	Milky Way	Solar System	Universe
☐ **B**	Milky Way	Universe	Solar System
☐ **C**	Solar System	Universe	Milky Way
☐ **D**	Solar System	Milky Way	Universe

(b) Some visible light telescopes are located in space.

Other visible light telescopes are located on the Earth's surface.

Explain why the images produced by telescopes on Earth are less clear than the images produced by telescopes in space.

(2)

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(c) This simplified diagram compares spectra of light from the Sun and two galaxies.

The light from galaxy 1 and galaxy 2 both show redshift.

Explain what these redshifts predict about the position and movement of the two galaxies.

(3)

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(d) Scientists have studied stars to discover how stars evolve.
They know that stars form in a nebula when clouds of dust and gas are pulled together by gravity.
Describe how this process continues for stars much more massive than the Sun.

(4)

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(Total for Question is 10 marks)

Mark Scheme

Q1.

Question Number	Answer	Additional guidance	Mark
(i)	Any one from:- to view distant objects /things far away (1) looking at planets /stars / space/ galaxies (1) bird watching (1)	 accept magnify distant objects accept other named astronomical objects	(1) AO2

Question Number	Answer	Mark
(ii)	**C** P and Q are both converging **A** is incorrect the lens Q is not diverging **B** is incorrect the lens P is not diverging **D** is incorrect neither lens is diverging	(1) AO3

Question Number	Answer	Additional guidance	Mark
(iii)	substitution (1) (power =) $\frac{1}{0.14}$ evaluation (1) 7.1 (D)	 7.14(28...) allow 7 award 1 mark for 7.1 to any other power of ten award 1 mark for correct answer as a fraction award full marks for the correct answer with no working	(2) AO2

Q2.

Question number	Answer	Additional guidance	Mark
	gravitational attraction	gravity gravitational field /force	**1** **AO1.1**

Q3.

	Answer	Additional guidance	Mark
(i)	gravitational attraction / gravitational force (causing collapse) (1)	allow gravity ignore weight ignore gpe ignore gravitational energy	**(1)** **AO1**

	Answer	Additional guidance	Mark
(ii)	An explanation linking: (gravity causing) increase in temperature (1) (until hot enough for) fusion (1) (until) balance (between gravity and fusion/thermal) (1)	 allow increase in pressure/density hydrogen to form helium allow nuclear reactions ignore fission allow equilibrium / counteracts	**(3)** **AO1**

Q4.

Question Number	Answer	Acceptable answers	Mark
(a)(i)	In ANY order smaller than the object (1) upside down (1)		**(2)**

Question Number	Answer	Acceptable answers	Mark
(a)(ii)	The idea that it can be captured on a screen	appears where the rays (of light) meet ignore references to reflection	**(1)**

Question Number	Answer	Acceptable answers	Mark
(a)(iii)	A (10)		**(1)**

Question Number	Answer	Acceptable answers	Mark
(a)(iv)	A (refraction by a converging lens)		**(1)**

Question Number	Answer	Acceptable answers	Mark
(b)	image was larger/magnified (compared with naked eye).	better {information / detail} / see {further / clearer /further away} / fainter /more objects / zoom in OWTTE	**(1)**

Question Number	Answer	Acceptable answers	Mark
(c)	Description to include: Geo = Earth at the centre (of the Solar System) (1) Helio = Sun at the centre (of the Solar System) (1)	"everything goes around the Earth / us" the Sun goes around the Earth "everything goes around the Sun" we revolve around the Sun Allow answers in diagram form Allow 1 mark for **both** descriptions the wrong way round	**(2)**

Q5.

	Answer	**Acceptable answers**	**Mark**
i	infrared (radiation)/(rays) (1)		**(1)**
ii	An explanation linking any two from • above the clouds / no clouds/ no weather (1) • image is clearer/more detailed/ not distorted/not blurred (1) • no light pollution (1) • (some) telescopes use gamma/ X-rays/ultraviolet /infrared/microwaves (1) • no absorption (by atmosphere) of gamma/ Xrays/ ultraviolet /infrared/ microwaves (1)	Credit to be given for stating that all telescopes would be better in space, but size and weight may exclude e.g. Jodrell Bank from space. no air/dust/pollution wider field of view/ can use anytime IG NORE 'see further' IGNORE 'it is closer (to the stars/planets)' IGNORE: references to improving understanding / knowledge of space	**(2)**

Q6.

Question number	Answer	Mark
(i)	(nuclear) fusion	(1)

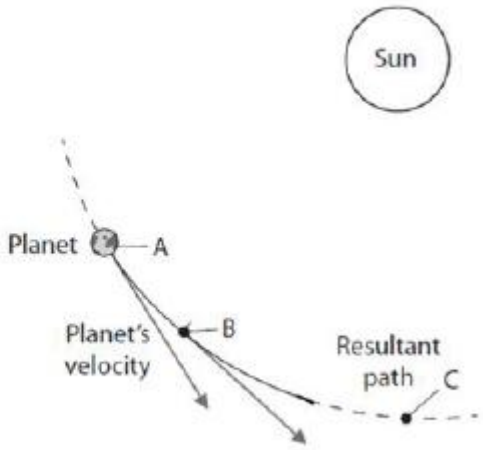
Question number	Answer	Additional guidance	Mark
(ii)	gravitational (pull)	gravity	(1)

Q7.

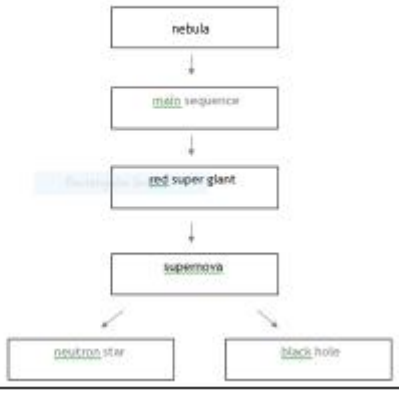
	Answer	Acceptable answers	Mark
(i)	D red giant then white dwarf		**(1)**
(ii)	an explanation linking **two** of the following: • Different/more wave(length)s/frequencies (now) detected/collected (1) • because telescopes positioned above Earth's atmosphere or by radio telescopes (1) OR • weaker signal(s) (now) detected/collected (1) • because modern telescopes are more powerful/have greater magnification **or** positioned above Earth's atmosphere/ on top of mountains (1) If no other marks awarded allow 1 mark for idea that: Electronic(s)/computers can process/improve the data/signal information/waves/radiation	named type of em radiation accept CMB for microwave space telescope or named space telescope (because) less or no light/radiation is absorbed by Earth's atmosphere Accept named space telescope eg Hubble/Planck/Compton etc Ignore references to clearer images/more detail/can see further/photographs	**(2)**

Q8.

Question number	Answer	Additional guidance	Mark
(i)	(force of) gravity/ gravitation		(1)

Question number	Answer	Additional guidance	Mark
(ii)	 Figure 2	Judge by eye. Horizontal line with arrow to the right from C	(1)

Q9.

Question number	Answer	Additional guidance	Mark
			(3)

Q10.

Question number	Answer	Mark
	C natural satellite	(1)

Q11.

	Answer	Acceptable answers	Mark
(a)	D		**(1)**
(b)	An explanation linking any two of (presence of Earth's) atmosphere (1) causes light to be absorbed/reduced in intensity (1) causes distortion of the image(1) (more) **light** pollution (1) (bigger) variations in temperature (1)	Accept reverse argument (more) air/ clouds/ pollution/ dust blocked / (more) difficult to see through blurs the image / refracts the light	**(2)**
(c)	An explanation linking any three of galaxies moving (1) away from Earth / Sun (1) galaxy 2 (moving away) faster (than galaxy 1) (1) galaxy 2 is (likely to be) most distant galaxy (1)	galaxies are (moving) at different speeds / away from each other / universe is expanding	**(3)**
(d)	A description including the following stages (up to 3 marks) Protostar (1) Main sequence star (1) (super) red giant (1) supernova (1) neutron star (1) (even more massive star can become) black hole (1) more massive stars have shorter life (1) Three stages in the correct sequence (1)		**(4)**